Chapter 15, Problem 5P

Problem

Find the Laplace transform of each of the following functions:

(a) $t2\cos(2t + 30°)u(t)$

(b) $3t4e{-}2t\ u(t)$

(c) $2tu(t) - 4\frac{d}{dt}\delta(t)$

(d) $2e{-}(t{-}1)\ u(t)$

(e) $5u(t/2)$

(f) $6e{-}t/3\ u(t)$

(g) $\frac{d^n}{dt^n}\delta(t)$

Step-by-step solution

Step 1 of 7

(a)

Consider the signal $x(t)=t^2\cos(2t+30°)u(t)$.

Simplify the signal $x(t)$.

$$\begin{aligned} x(t) &= t^2(\cos 2t\cos 30° - \sin 2t\sin 30°)u(t) \\ &= t^2\left(\frac{\sqrt{3}}{2}\cos 2t - \frac{1}{2}\sin 2t\right)u(t) \\ &= \frac{t^2}{2}\left(\sqrt{3}\cos 2t - \sin 2t\right)u(t) \end{aligned}$$

Write the Laplace transform pairs that are used to find $X(s)$.

$$\cos atu(t) \xleftrightarrow{L} \frac{s}{s^2+a^2}$$

$$\sin atu(t) \xleftrightarrow{L} \frac{a}{s^2+a^2}$$

$$t^n f(t) \xleftrightarrow{L} (-1)^n \frac{d^n F(s)}{ds^n}$$

Apply the Laplace transform pairs to find $X(s)$.

$$\cos 2tu(t) \xleftrightarrow{L} \frac{s}{s^2+2^2}$$

$$\sin 2tu(t) \xleftrightarrow{L} \frac{2}{s^2+2^2}$$

$$\begin{aligned} X(s) &= \frac{1}{2}\frac{d^2}{ds^2}\left(\sqrt{3}\frac{s}{s^2+2^2} - \frac{2}{s^2+2^2}\right) \\ &= \frac{1}{2}\frac{d}{ds}\left(\frac{(s^2+2^2)\sqrt{3} - (\sqrt{3}s-2)(2s)}{(s^2+2^2)^2}\right) \\ &= \frac{1}{2}\frac{d}{ds}\left(\frac{-\sqrt{3}s^2+4s+4\sqrt{3}}{(s^2+2^2)^2}\right) \end{aligned}$$

$$\begin{aligned} X(s) &= \frac{1}{2}\left[\frac{(s^2+2^2)^2(-2\sqrt{3}s+4) - 2(-\sqrt{3}s^2+4s+4\sqrt{3})(s^2+2^2)(2s)}{(s^2+2^2)^4}\right] \\ &= \left[\frac{\sqrt{3}s^3-6s^2-12\sqrt{3}s+8}{(s^2+4)^3}\right] \end{aligned}$$

Therefore, the Laplace transform $X(s)$ is $\boxed{\frac{\sqrt{3}s^3-6s^2-12\sqrt{3}s+8}{(s^2+4)^3}}$.

Step 2 of 7

(b)

Consider the signal $x(t)=3t^4e^{-2t}u(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$t^n f(t) \xleftrightarrow{L} (-1)^n \frac{d^n F(s)}{ds^n}$$

$$e^{-at}u(t) \xleftrightarrow{L} \frac{1}{s+a}$$

Apply the Laplace transform pairs to find $X(s)$.

$$e^{-2t}u(t) \xleftrightarrow{L} \frac{1}{(s+2)}$$

$$t^4 f(t) \xleftrightarrow{L} (-1)^4 \frac{d^4 F(s)}{ds^4}$$

$$\begin{aligned} X(s) &= 3\frac{d^4}{ds^4}\frac{1}{s+2} \\ &= \frac{72}{(s+2)^5} \end{aligned}$$

Therefore, the Laplace transform $X(s)$ is $\boxed{\frac{72}{(s+2)^5}}$.

Step 3 of 7

(c)

Consider the signal $x(t)=2tu(t)-4\frac{d}{dt}\delta(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$t^n f(t) \xleftrightarrow{L} (-1)^n \frac{d^n F(s)}{ds^n}$$

$$\frac{df}{dt} \xleftrightarrow{L} sF(s)-f(0^-)$$

$$\begin{aligned} X(s) &= 2tu(t)-4\frac{d}{dt}\delta(t) \\ &= \frac{2}{s^2}-4(s(1)) \\ &= \frac{2}{s^2}-4s \end{aligned}$$

Therefore, the Laplace transform $X(s)$ is $\boxed{\frac{2}{s^2}-4s}$.

Step 4 of 7

(d)

Consider the signal $x(t)=2e^{-(t-1)}u(t)$.

Rewrite the signal $x(t)$.

$$x(t)=e2e^{-t}u(t)$$

Write the Laplace transform pairs that are used to find $X(s)$.

$$e^{-at}u(t)\xleftrightarrow{L}\frac{1}{s+a}$$

Apply the Laplace transform pairs to find $X(s)$.

$$e^{-t}u(t)\xleftrightarrow{L}\frac{1}{s+1}$$

$$X(s)=\frac{2e}{s+1}$$

Therefore, the Laplace transform $X(s)$ is $\boxed{\frac{2e}{s+1}}$.

Step 5 of 7

(e)

Consider the signal $x(t)=5u\left(\frac{t}{2}\right)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$f(at)\xleftrightarrow{L}\frac{1}{a}F\left(\frac{s}{a}\right)$$

Apply the Laplace transform pairs to find $X(s)$.

$$u\left(\frac{t}{2}\right)\xleftrightarrow{L}2\left(\frac{1}{2s}\right)$$

$$X(s)=\frac{5}{s}$$

Therefore, the Laplace transform $X(s)$ is $\boxed{\frac{5}{s}}$.

Step 6 of 7

(f)

Consider the signal $x(t)=6e^{\left(-\frac{t}{3}\right)}u(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$e^{-at}u(t)\overset{\mathrm{L}}{\longleftrightarrow}\frac{1}{s+a}$$

Apply the Laplace transform pairs to find $X(s)$.

$$e^{-\frac{1}{3}t}u(t)\overset{\mathrm{L}}{\longleftrightarrow}\frac{1}{\left(s+\frac{1}{3}\right)}$$

$$\begin{aligned}X(s)&=\frac{6}{\left(s+\frac{1}{3}\right)}\\&=\frac{18}{3s+1}\end{aligned}$$

Therefore, the Laplace transform $X(s)$ is $\boxed{\frac{18}{3s+1}}$.

Step 7 of 7

(g)

Consider the signal $x(t)=\frac{d^n}{dt^n}\delta(t)$.

Write the Laplace transform pairs that are used to find $X(s)$.

$$\frac{d^n}{dt^n}f(t)\overset{\mathrm{L}}{\longleftrightarrow}s^nF(s)$$

Apply the Laplace transform pairs to find $X(s)$.

$$\frac{d^n}{dt^n}\delta(t)\overset{\mathrm{L}}{\longleftrightarrow}s^n(1)$$

$$X(s)=s^n$$

Therefore, the Laplace transform $X(s)$ is $\boxed{s^n}$.